UBT616: PROTEIN ENGINEERING

L	T	P	Cr
3	0	0	3.0

Course Objective: Aiming to provide basic knowledge of engineering and design of the protein for its application, this course will make students learn structural and functional relationships in proteins and enabling students to improvise protein structure and function.

Syllabus

Elements of protein structure: Introduction to protein engineering; Primary structure: amino acids and their –R groups; Secondary structure: α helix, β strand, β sheet, loops, Ramachandran plot; Supersecondary and tertiary structure: motifs, domain, and fold; Quaternary structure: oligomer assembly; Relationship between structure and function: protein active site, catalytic site, crypto sites and druggability, cooperativity and allosteric effect.

Experimental and computational tools used in protein structural Biology: Protein structure determination by X ray diffraction (XRD) and NMR, Prediction of protein structure and conformation from sequence data (Homology Modeling, Threading and *de novo* prediction); Computational tools for prediction of protein active sites; Spectroscopy methods (CD and fluorescence) of determination of protein structural conformation; Protein activity and stability measurement (k_{cat}/K_m; T_m) using spectroscopy.

Protein Engineering: Mutagenesis methods: site directed mutagenesis–insertion, deletion, substitution, modular protein domain, random mutagenesis- directed evolution, gene shuffling; Kunkle mutagenesis; Phage display technology; Overview of CRISPR/Cas method (*in vivo*); Insertion of unnatural amino acids in protein using orthogonal system; Chemical modifications of proteins.

Protein expression and purification systems: Expression of proteins in bacteria, yeast, insect and mammalian cells; Protein purification overview; Different chromatography methods in protein purification (Affinity, ion exchange, gel exclusion, hydrophobic).

Application of Protein Engineering: Case study – protein engineering in lysozyme; Overview of protein design and considerations; Applications in drug delivery, biosensors, immunotherapy; antibody engineering.

Course Learning Objectives (CLO)
The students will be able to:

1. Understand the protein structure and function relationship.
2. Know the methods for determination of protein structure and studying protein structure, function and stability.
3. Apply the methods involved in protein engineering.

4. Design strategy of protein expression and purification.
5. Comprehend the scope of protein engineering application.

Text Books

1. Primrose SB and Twyman RM: Principles of Gene Manipulation and Genomics Blackwell Publishing (2006).
2. Cleland JL and Craik CS: Protein Engineering: Principles and Practice, Wiley-Liss. (1996).
3. Lutz S and Bornscheuer U T: Protein Engineering Handbook, Wiley-VCH (2009).
4. Zhao H(Editor), Lee SY(Series Editor), Nielsen J(Series Editor), Stephanopoulos G (Series Editor): Protein Engineering: Tools and Applications, Wiley-VCH (2021).
5. Park SJ(Editor), Cochran JR(Editor), Protein Engineering and Design, CRC Press (2009).

Reference Books

1. Branden CI, Tooze J: Introduction to Protein Structure, Garland Science (1998).
2. Williamson M: How Proteins Work, Garland Science (2012).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30